AI Basketball Practice Coach

Project plan & implementation status report

22 October 2026 | Repository: Innovative-Design-Project/pose-estimation (main @ 66f6d98)

1. Summary

The product is a camera-based coach for **basketball practice sessions**. A single camera on a tripod watches one player. It offers two modes: **(1) live voice coaching**, where shot-by-shot feedback is spoken through ElevenLabs, and **(2) post-session analytics**, a dashboard to review the session afterwards. Both modes use the same core: pose estimation, ball and hoop tracking, event detection (shots, dribbles, layups) and biomechanical metrics.

Where we are: the repository has a working pose-estimation foundation (17-keypoint YOLO inference on images and video) and a complete tooling pipeline for fine-tuning and evaluating the pose model. **None of the product features exist yet**: no live camera loop, ball or hoop detection, shot detection, metrics, voice output or analytics. We estimate the foundation is roughly the first 10-15% of the total work.

Three recommendations before writing more code

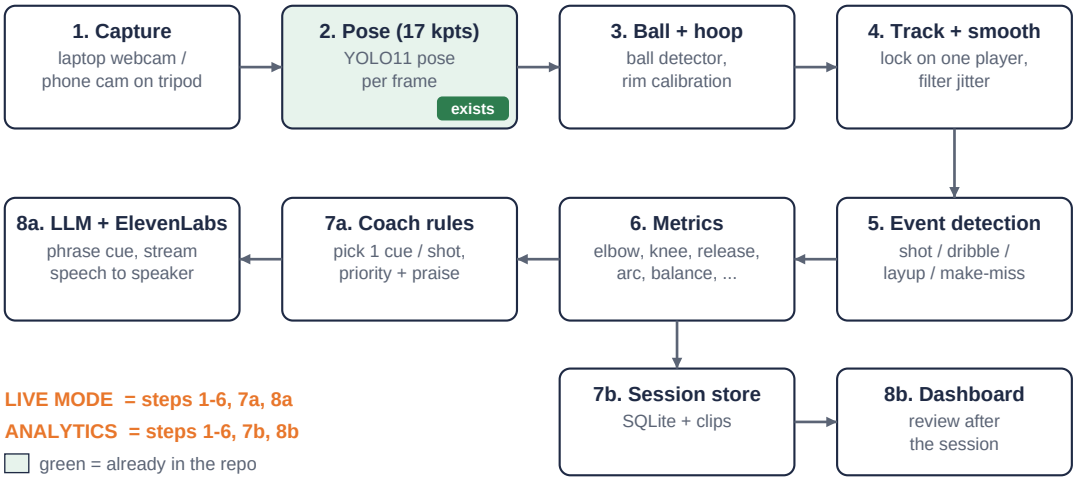
1. **Record our own practice footage now.** The current dataset is 8,595 mixed broadcast/game frames with many players, while the product sees one player practicing in a gym. Footage from the real camera setup is needed both to build features and to measure accuracy.
2. **Build the shot pipeline on the off-the-shelf pose model first.** A single, unoccluded player is the easy case for YOLO pose. Fine-tune only if evaluation on our own footage shows wrist/elbow errors that hurt the metrics.
3. **Ball and hoop detection is the critical path.** Make/miss, shot arc, dribble counting and release timing all depend on it, and nothing for it exists yet.

2. Confirmed decisions

Topic	Decision
Voice	ElevenLabs text-to-speech, streamed to the laptop speaker / earbuds
Platform	Python app on a laptop; webcam or phone camera on a tripod; analytics as a web dashboard
v1 activities	Shooting form, make/miss tracking, dribbling, layups/footwork
Feedback engine	Rule-based metrics computed from keypoints + ball track; an LLM turns the top issue into a short spoken cue
Users	One player at a time (assumed; see open questions)

3. Target architecture

Both modes share steps 1-6. Live mode adds coaching rules and voice, and analytics mode adds storage and a dashboard. Recording every session to the store, including live sessions, means analytics come for free after each live session.



Measured performance on the team laptop (Apple M4, 720p frames, pose only)

Model	Device	Latency / frame	Throughput	Fit for live mode
yolo11n-pose	CPU	20 ms	~51 fps	Yes - leaves headroom for ball detector
yolo11n-pose	MPS (GPU)	22 ms	~45 fps	Yes
yolo11m-pose	MPS (GPU)	52 ms	~19 fps	Borderline once ball detection is added
yolo11m-pose	CPU	80 ms	~12 fps	No

Live mode needs about 25-30 fps for pose plus ball detection, so the likely choice is the n or s model on MPS, with the m model reserved for offline analytics reprocessing. Note: the shooting release lasts ~0.1-0.2 s, so the camera should record at 60 fps if possible, even when only every second frame is analysed live.

4. What is implemented today

Component	Where	Status	Notes
Pose inference on images and videos, annotated output	main.py	**Done**	Batch over images/ and videos/ folders; auto device (CUDA/MPS/CPU). File input only, no live camera.
Default model weights	main.py, README	**Partial**	Defaults to *yolo26m-pose.pt*, which is not in the repo; the pipeline uses yolo11m-pose.pt. Needs one consistent choice.
Frame pool download (Roboflow)	sports_pipeline/download_frames.py	**Done**	8,595 basketball frames with Ball / Player / Referee boxes. Mostly game/broadcast footage, not practice.
Pseudo-labelling with base model	pseudo_label.py	**Done**	6,522 frames have pose pseudo-labels. Still lists hockey.
CVAT correction round-trip	to/from_coco_keypoints.py	**Done**	Prefilled skeletons out to CVAT and corrected labels back; supports gold val/test sets.
Clip-level train/val/test split	split_dataset.py	**Done**	251 test / 250 val / 8,089 train frames, whole clips per split (no leakage).
Dataset build + fine-tune script	build_dataset.py, train.py	**Built, not run**	No fine-tuning run has been done yet.
Evaluation script (pose mAP)	evaluate.py	**Built, not run**	Tested end to end on pseudo labels only; the real baseline needs hand-labelled test frames.
Labelling guide	LABELING.md	**Done**	Workflow and labelling rules.
Docs / research	README, YOLO report, architecture diagram	**Partial**	README still describes yoga/dance; YOLO report is general model research.

5. What we need that is not implemented

Priority: **P0** = needed for any demo, **P1** = needed for v1, **P2** = nice to have. Effort is relative (S = days, M = 1-2 weeks, L = several weeks) for one developer.

Module	What it does	Pri.	Effort
Live capture loop	Read webcam / phone stream (e.g. via Continuity Camera or IP cam), run models per frame, show overlay, record raw video	P0	S
Practice footage dataset	10-20 recorded sessions with several players, left/right handers, makes and misses, from the agreed camera position	P0	M
Player lock + smoothing	Pick the practising player (ignore people in background); smooth keypoints over time (e.g. One-Euro filter)	P0	S
Ball detector	YOLO detect model for the ball, trained on Roboflow ball labels + our footage; ball track with gap filling	P0	M
Hoop / rim location	One-click rim calibration at session start (fixed camera); optional rim detector later	P0	S
Shot detection	State machine: gather, set point, release, follow-through, landing (from wrist/ball height and ball leaving hand)	P0	M
Make / miss	Ball passes down through the rim region = make; else miss; plus attempt count	P0	M
Shooting form metrics	Elbow angle at set point, knee bend, release height & timing, arc angle, follow-through hold, balance / landing drift	P0	M
Coaching rules	Rank issues against target ranges, choose one cue per shot, add praise, avoid repeating the same cue	P0	S
LLM cue phrasing	Turn the chosen issue + numbers into a short (<12 words) natural cue; strictly grounded in the metrics	P1	S

Module	What it does	Pri.	Effort
ElevenLabs voice	Streaming TTS, cache of common cues, mute/volume, never speak mid-shot	P0	S
Session store	SQLite: sessions, shots, events, metrics; short video clip per shot	P1	S
Analytics dashboard	Web dashboard (e.g. Streamlit): make %, form trends, consistency, per-shot replay with skeleton, LLM session summary	P1	M
Dribbling module	Bounce count and cadence, dribble height vs. knee/hip, hand used, crossover detection	P1	M
Layup module	Steps between gather and takeoff, takeoff foot, finishing hand, approach path	P1	M
Shot location / zones	Map the player's feet to court distance bands via a one-time court calibration	P2	M
Accuracy evaluation	Hand-label ~200 practice frames + manually counted shots to measure pose, shot-detection and make/miss accuracy	P0	S
Pose fine-tuning	Only if the evaluation shows it is needed; reuse the existing pipeline on practice footage	P2	M
Settings & onboarding	Camera placement guide, handedness, drill selection, voice on/off	P1	S
Tests	Unit tests for metric maths and event detection on recorded clips	P1	S

6. What we do not need (or can defer)

Item	Decision	Reason
Hockey code paths and data	Remove	Project is basketball only.
Yoga / dance samples and README text	Remove / rewrite	Already moved to assets-legacy/; README still describes them.
Multi-player broadcast tracking (SportsMOT-style), referee class	Drop	Practice sessions have one player; broadcast games are a different problem.
Fine-tuning on the current broadcast frame pool	Defer	Wrong domain for practice coaching. Keep the tooling and run it on our own footage if needed.
Large pose models (yolo11x / l) for live use	Drop for live	Too slow on the laptop for live; can be used for offline reprocessing.
Trained ML shot-quality classifier	Defer to v2	Needs a labelled good/bad shot dataset we do not have; rules + LLM cover v1.
3D pose / multi-camera	Defer to v2	A fixed single view with known placement gives usable 2D angles.
Mobile app, cloud backend, user accounts	Not in v1	Laptop app was chosen; analytics dashboard runs locally.
Further YOLO generation comparisons	Stop	The model choice is made; spend the time on product features.

7. Metrics per activity (v1)

All metrics are computed from 2D keypoints and the ball track, so they depend on a fixed, known camera position (see section 9). Target ranges should be set with a coach and compared against each player's own best shots, not one universal 'perfect form'.

Activity	Metric	How it is computed
Shooting	Elbow angle at set point	Shoulder-elbow-wrist angle when the ball is at its highest point before release
	Knee bend depth	Minimum hip-knee-ankle angle during the dip
	Release height & timing	Wrist height (relative to head) at release; delay between leg extension and release

Activity	Metric	How it is computed
	Shot arc	Launch angle of the ball trajectory fitted over the first frames after release
	Follow-through	Time the shooting wrist stays above the head after release
	Balance / landing	Horizontal drift of ankles between takeoff and landing; torso lean (shoulder-hip line)
Make/miss	Result, attempts, FG%	Ball crosses the calibrated rim region from above and exits below
	Streaks, % by distance band	From the shot log; distance bands need the court calibration (P2)
Dribbling	Count & cadence	Ball height minima (bounces) per second
	Dribble height	Ball apex relative to the player's hip / knee
	Hand usage & crossovers	Closest wrist to the ball at each apex; hand switches
	Head up (approx.)	Face keypoint visibility and head orientation; rough in 2D, so labelled approximate
Layups	Step count & takeoff foot	Ankle contact events between gather and takeoff
	Finishing hand, jump height proxy	Wrist at release; hip rise relative to standing height

8. Live voice coaching loop

The cue must arrive while the player is still thinking about the shot, but never during the next one. Target: **under ~2 seconds from landing to first spoken word**.

Step	Budget	Notes
Shot end detected + metrics	~0.1 s	Local, from the last ~2 s of buffered frames
Coaching rule picks one cue	~0 s	Local; priority order, praise for good shots, no repeats
LLM phrases the cue	~0.5-1 s	Small fast model (e.g. Claude Haiku class); prompt contains only the metrics
ElevenLabs streaming TTS	~0.3-0.5 s	Low-latency Flash voice model, streamed playback starts on first audio chunk
Fallback	instant	Pre-generated audio for the ~30 most common cues; used offline or when slow

Example: metrics say elbow flares 18 degrees out and arc is 38 degrees (flat). The rule ranks the flat arc first. The LLM says: *"Flat shot - push up through the ball, aim higher."* ElevenLabs speaks it.

Post-session analytics (v1 contents)

- Session overview: attempts, makes, FG%, session length, best streak.
- Form trends: each metric per shot over the session (fatigue shows as drift late in the session).
- Consistency: spread of release angle, elbow angle and arc, which matters more than any single value.
- Shot list with replay: short clip per shot with skeleton and ball path overlay, filterable by make/miss.
- Session-to-session comparison and an LLM-written summary with 2-3 drills for next time.
- Export: CSV of the shot log and a PDF session report.

9. Roadmap

Phase	Deliverables	Done when
0. Foundations	Fix model default; live capture loop with fps overlay and recording; agree camera placement (side / 45 degrees on the shooting-hand side, ~4-6 m, whole body and rim in frame, 60 fps); record first practice sessions	Live skeleton at 25+ fps on the laptop; 10+ sessions recorded
1. Perception	Player lock + smoothing; ball detector; rim calibration; hand-label ~200 practice frames and evaluate pose with evaluate.py	Ball tracked through shots; pose accuracy measured on our footage
2. Shooting	Shot detection, make/miss, shooting form metrics	Proposed targets: >=95% of shots detected, >=90% make/miss correct vs manual count
3. Live coaching	Coaching rules, LLM phrasing, ElevenLabs streaming, cue cache, mute	Cue spoken <2 s after landing in a real session
4. Analytics	Session store, dashboard, per-shot replay, LLM summary, export	Full session reviewable after practice
5. Dribbling & layups	Dribble and layup detectors and metrics, added to live and analytics	Counts match manual counts on test clips
6. Validation	Sessions with players and a coach; tune thresholds; fine-tune pose only if needed	Coach agrees with most cues

10. Risks and mitigations

Risk	Mitigation
2D angles change with camera angle	Fixed placement guide + on-screen check that the whole body and rim are visible; compare to the player's own baseline
Shooting arm hidden behind the body	Camera on the shooting-hand side; handedness setting
Motion blur at release; small fast ball	60 fps capture, good lighting, ball detector trained on our footage
Internet needed for LLM and ElevenLabs	Cached cue audio and template phrasing as offline fallback
LLM gives wrong or invented advice	LLM only rephrases the issue the rules chose; short output limit; no free-form coaching
API cost per session	One short cue per shot, caching repeated cues; estimate cost per hour during Phase 3
Privacy of recorded video	Store locally, ask consent (especially minors), allow deleting sessions

11. Open questions for the team

- **Team size and deadline:** is there a submission or demo date? This decides how much of Phase 5 fits in v1.
- **Where do sessions happen?** Indoor gym or outdoor court, and how reliable is the internet there?
- **Left-handed players:** support from v1? (Recommended: yes, it is a mirror setting.)
- **Coach involvement:** who defines the target ranges for 'good form'? Can a coach review cues during Phase 6?
- **Voice:** language(s), voice persona, and how often to speak (every shot vs. every few shots)?
- **Budget:** monthly limit for ElevenLabs and LLM API usage?
- **Drills:** free shooting only, or structured drills (e.g. 10 shots from 5 spots) that the app guides?